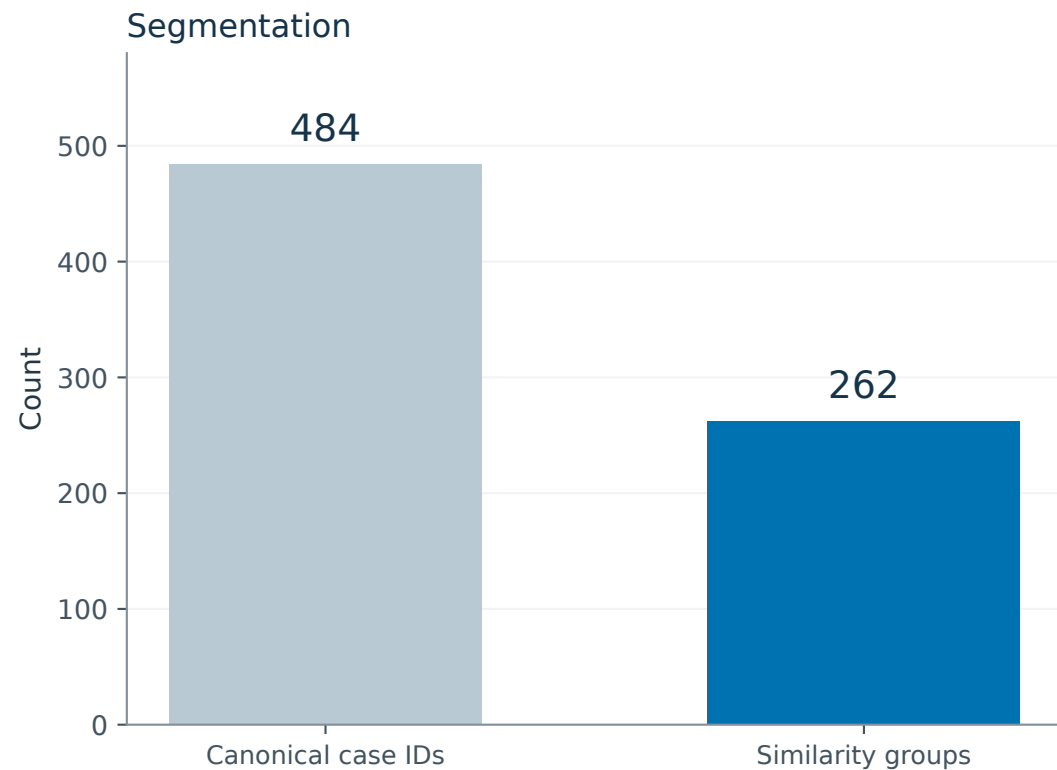
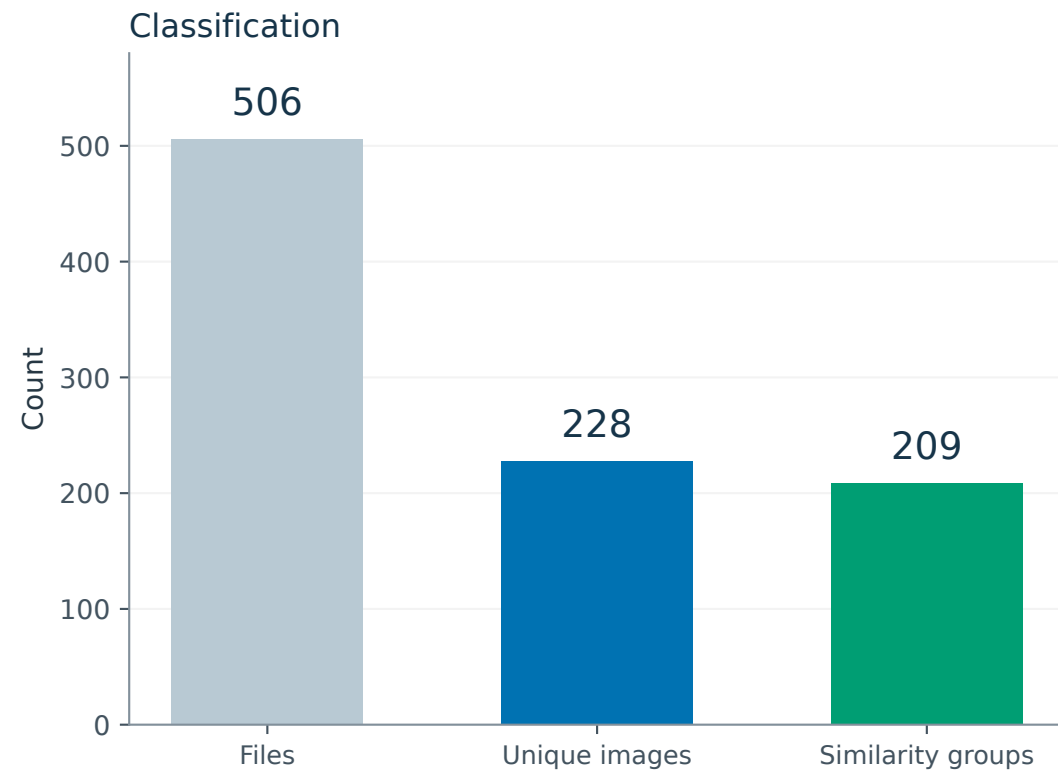


# 01 | Audit the sampling unit before measuring performance

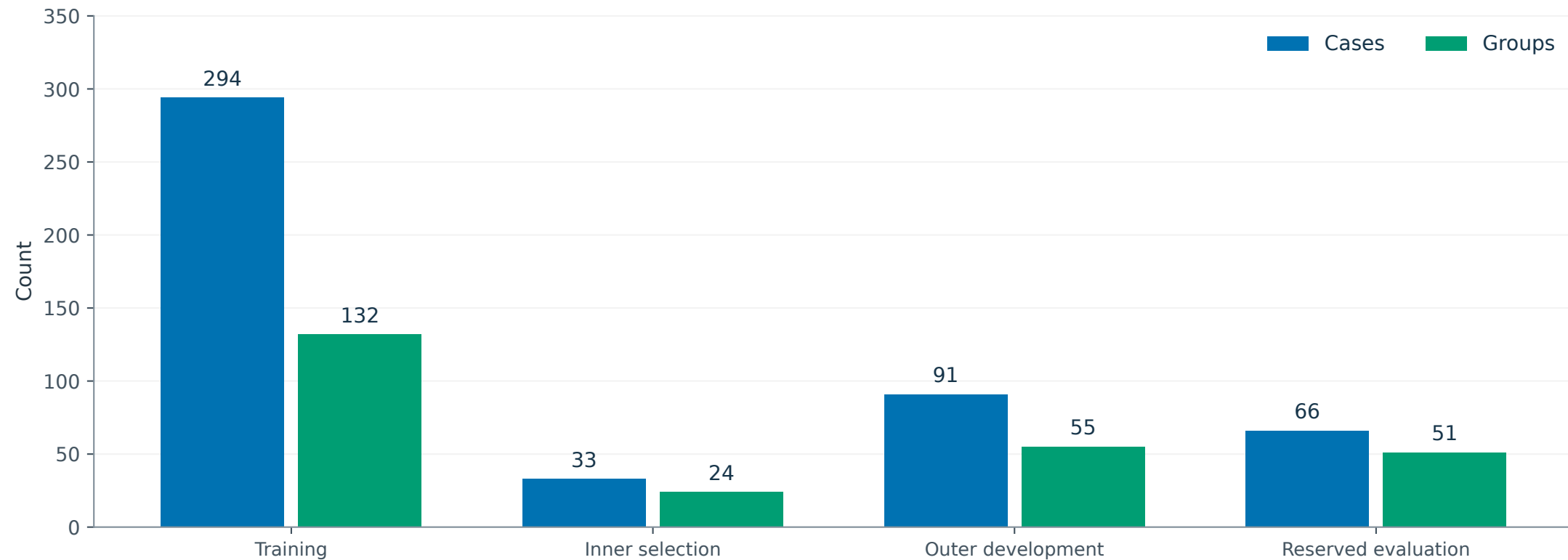
File counts and case IDs overstate the number of independent image groups



Source: completed dataset audits. Similarity groups are not verified patients.

## 02 | Frozen segmentation cohorts

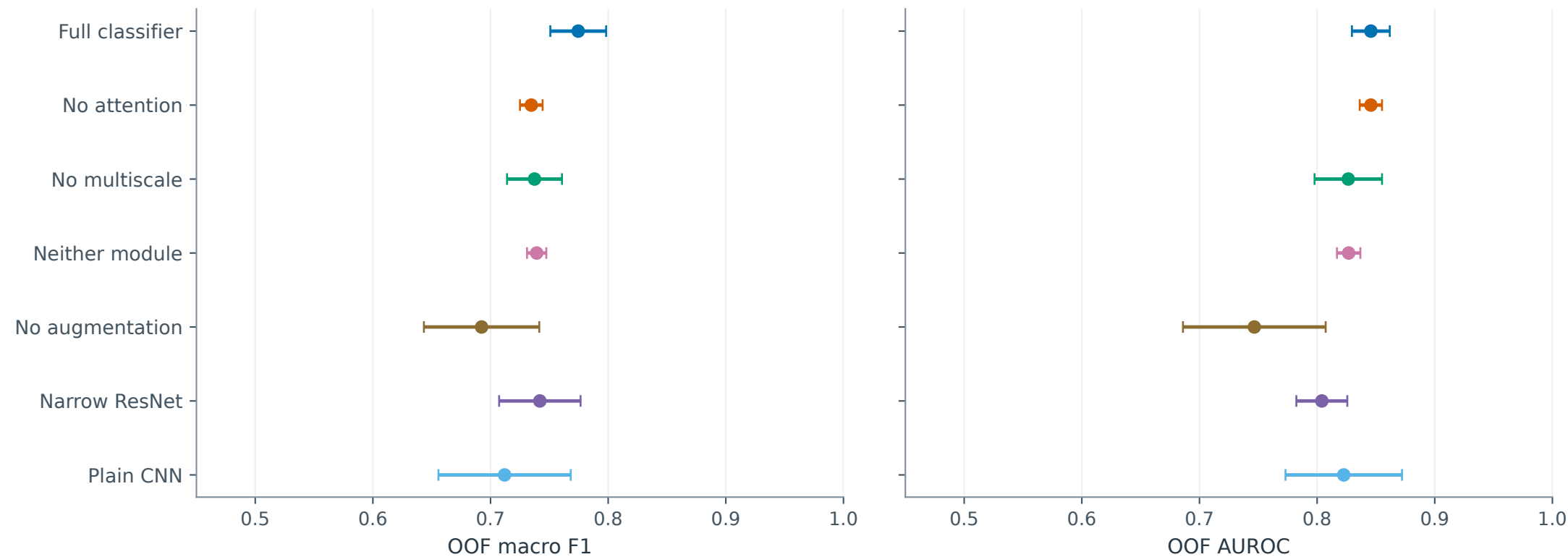
Only development fold 0 was trained; three seeds repeat the same partition



Historical exposure remains: seven reserved groups contain earlier notebook validation cases.

### 03 | Classification ablations across three seeds

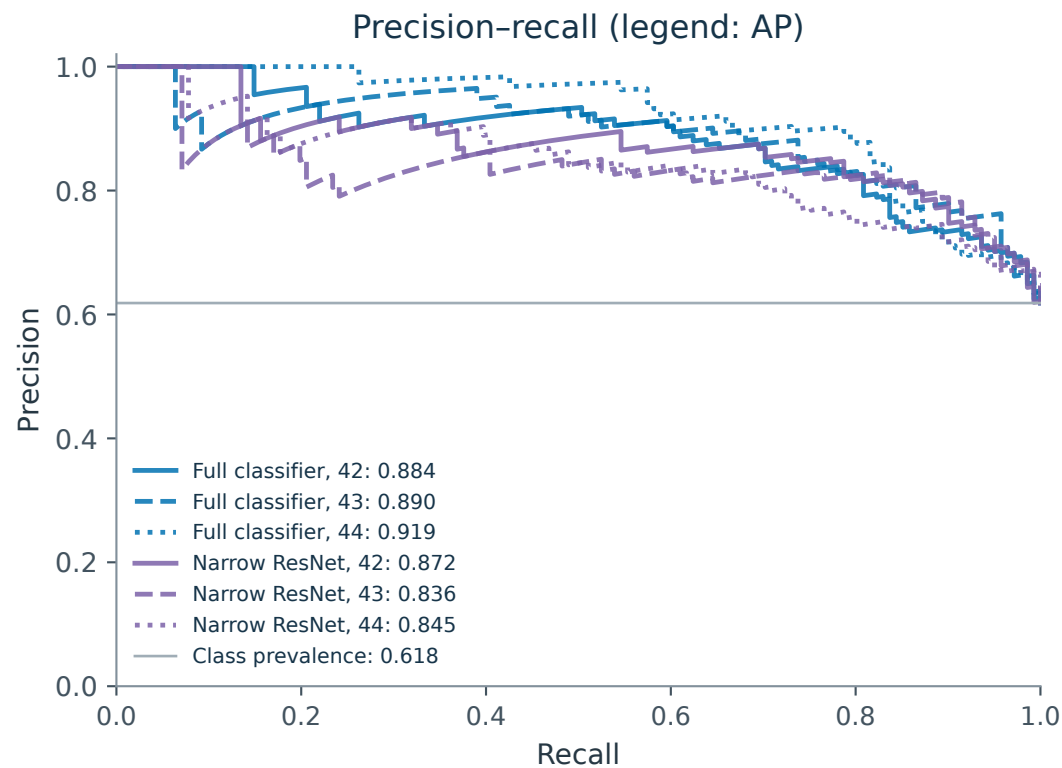
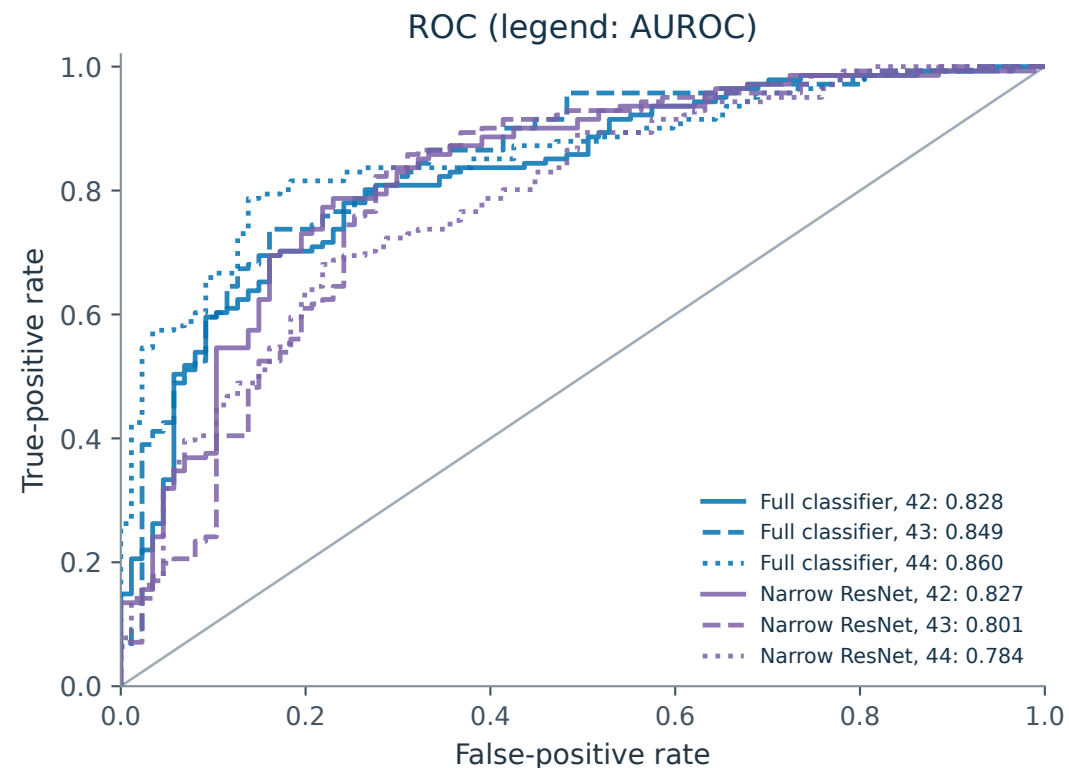
Seven variants × five grouped folds × three seeds = 105 completed fits



228 unique images; 209 conservative groups. Bars:  $\pm 1$  seed SD, not patient uncertainty.

## 04 | Out-of-fold discrimination

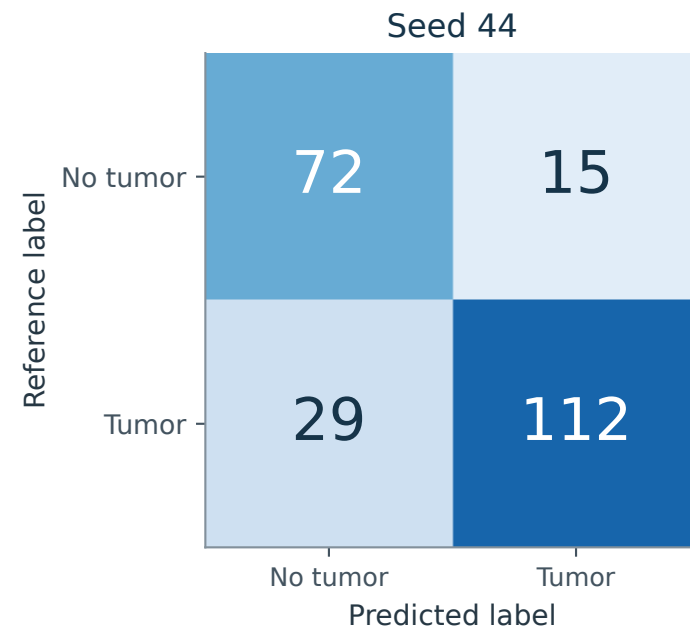
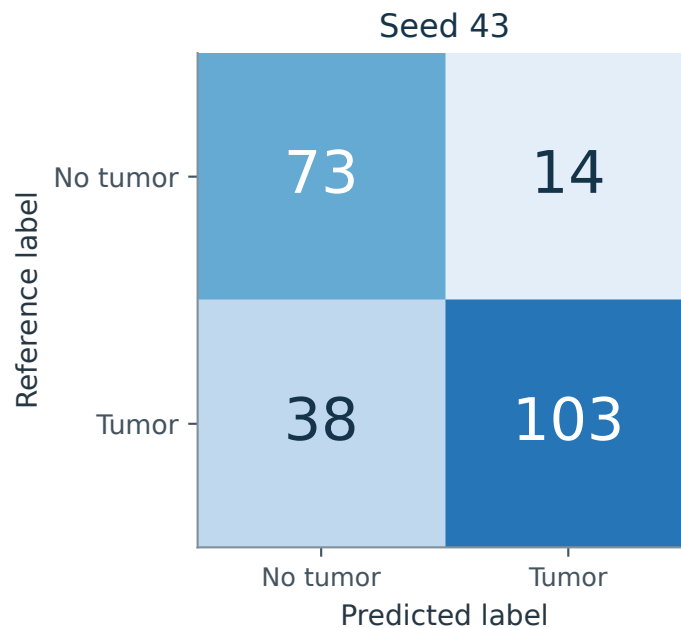
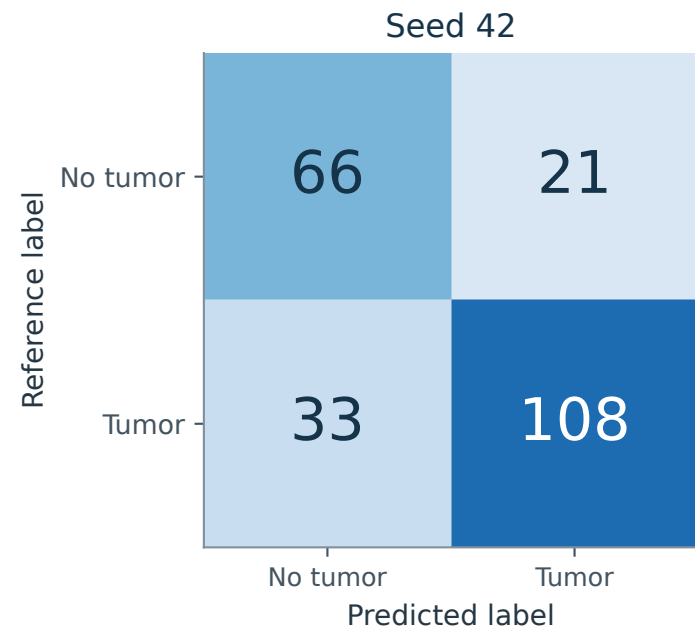
Full classifier and narrow ResNet; a separate curve for each fixed seed



No new training, threshold selection or confidence band. Five grouped folds contribute to each seed curve.

## 05 | Classification errors at the frozen 0.5 threshold

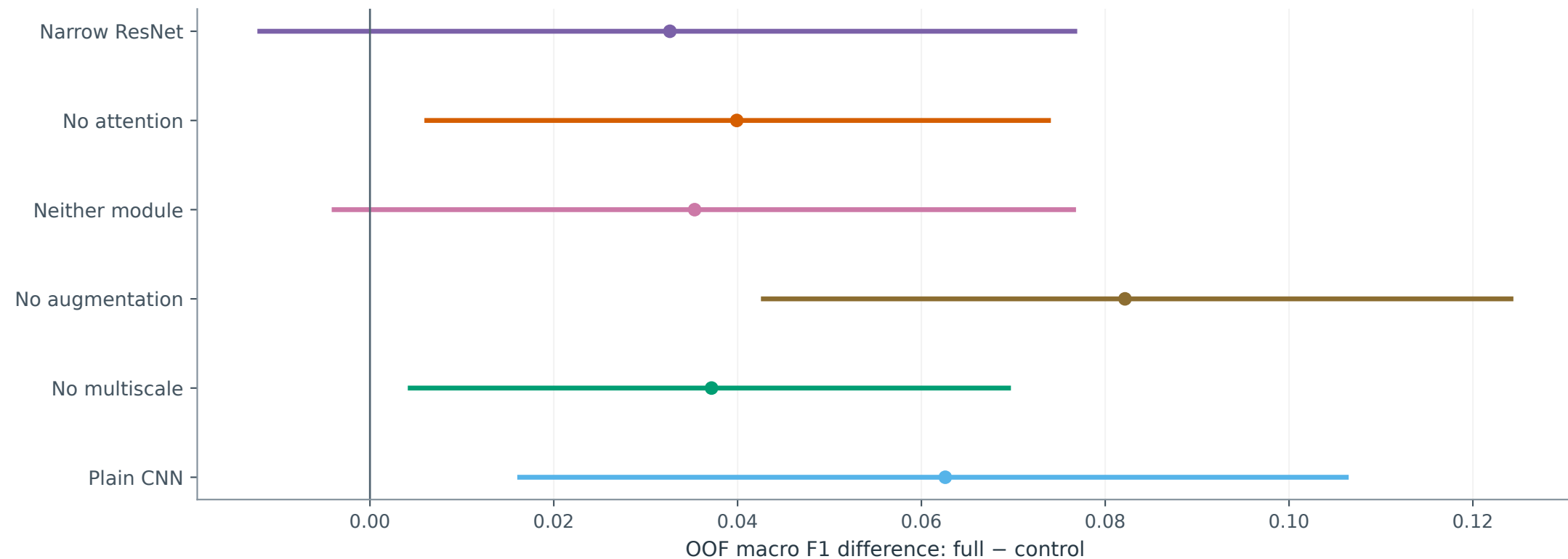
Counts are shown independently for each seed; positive label = tumor image



Threshold fixed at 0.5; no post hoc operating-point optimization.

## 06 | Paired classification contrasts

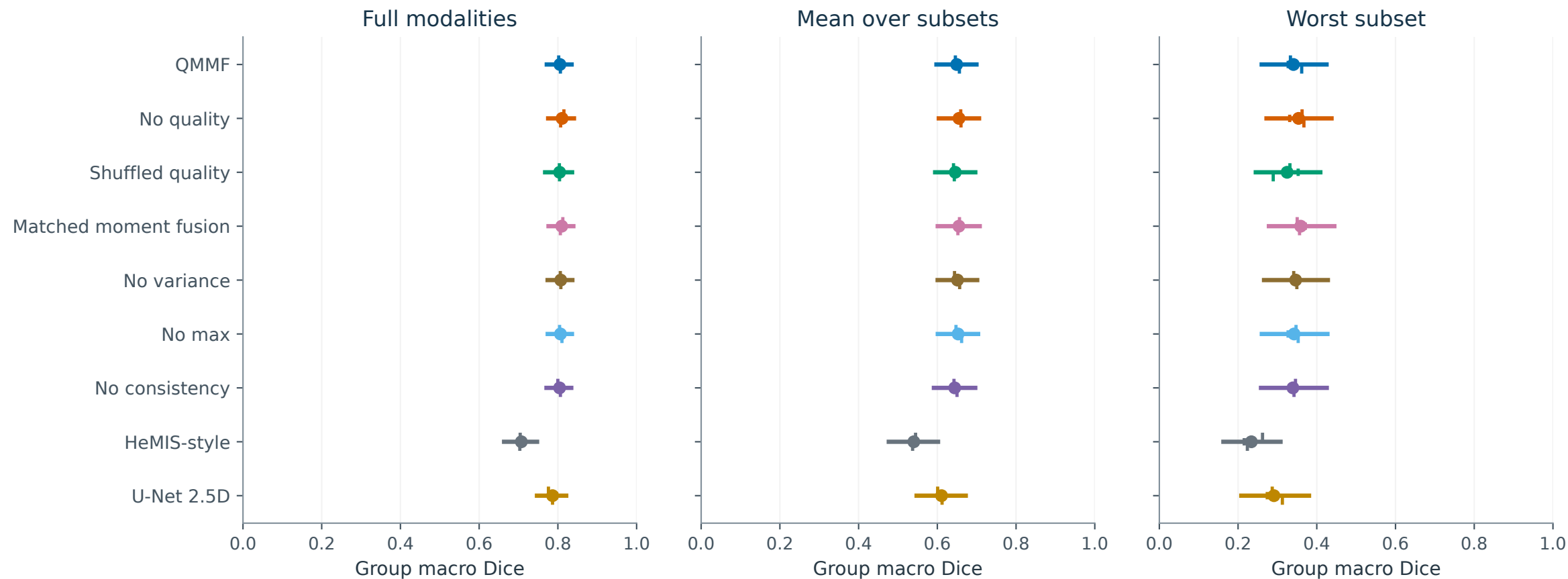
Full classifier minus each control; descriptive group-bootstrap intervals



Group bootstrap; fixed training seeds. Unadjusted descriptive intervals; no new hypothesis tests.

## 07 | Main segmentation: all nine variants

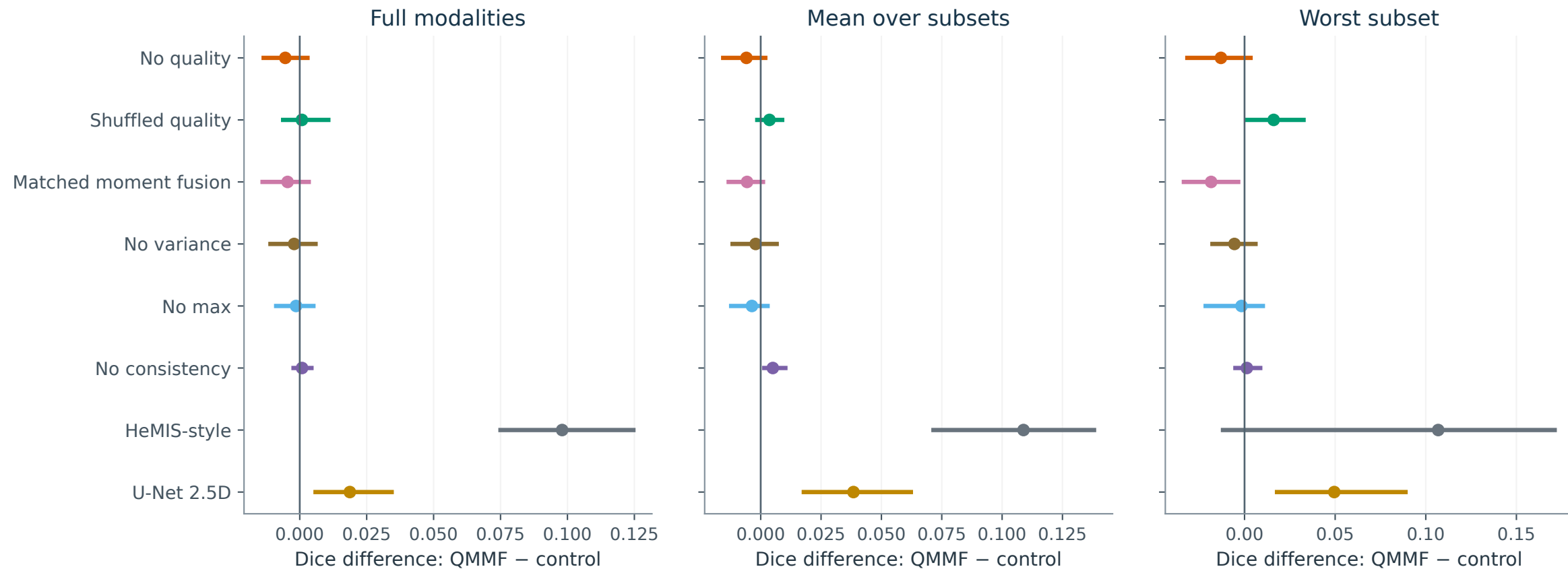
27 completed fits; full and missing-modality development endpoints



Bars: 95% group-bootstrap intervals, fixed seeds. Ticks: seeds 42/43/44. Groups are not verified patients.

## 08 | Main paired ablation intervals

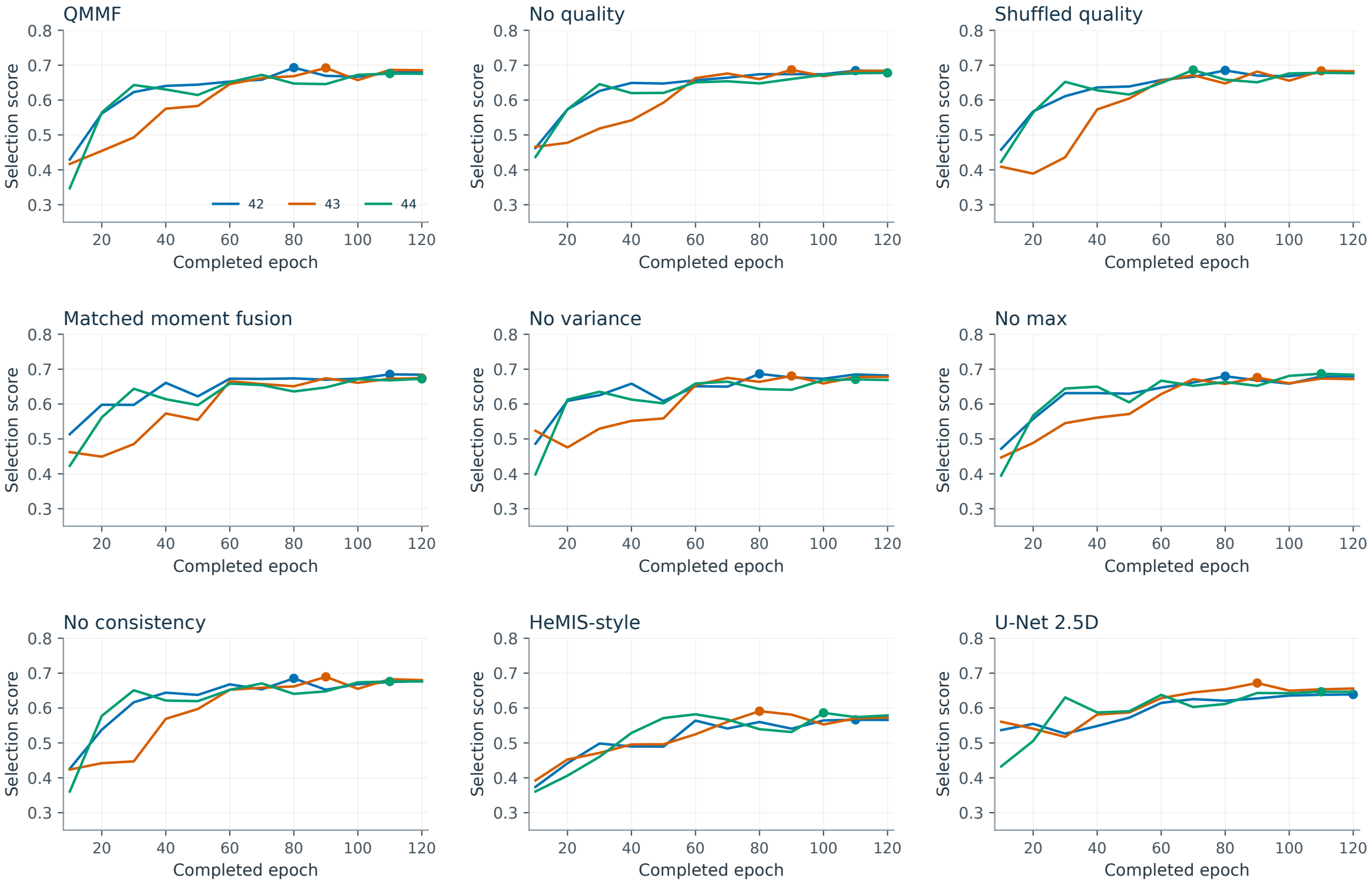
QMMF minus control; Bonferroni intervals for 8 controls per endpoint



Paired resampling unit: conservative image group. Adjustment is within endpoint, not across the entire figure gallery.

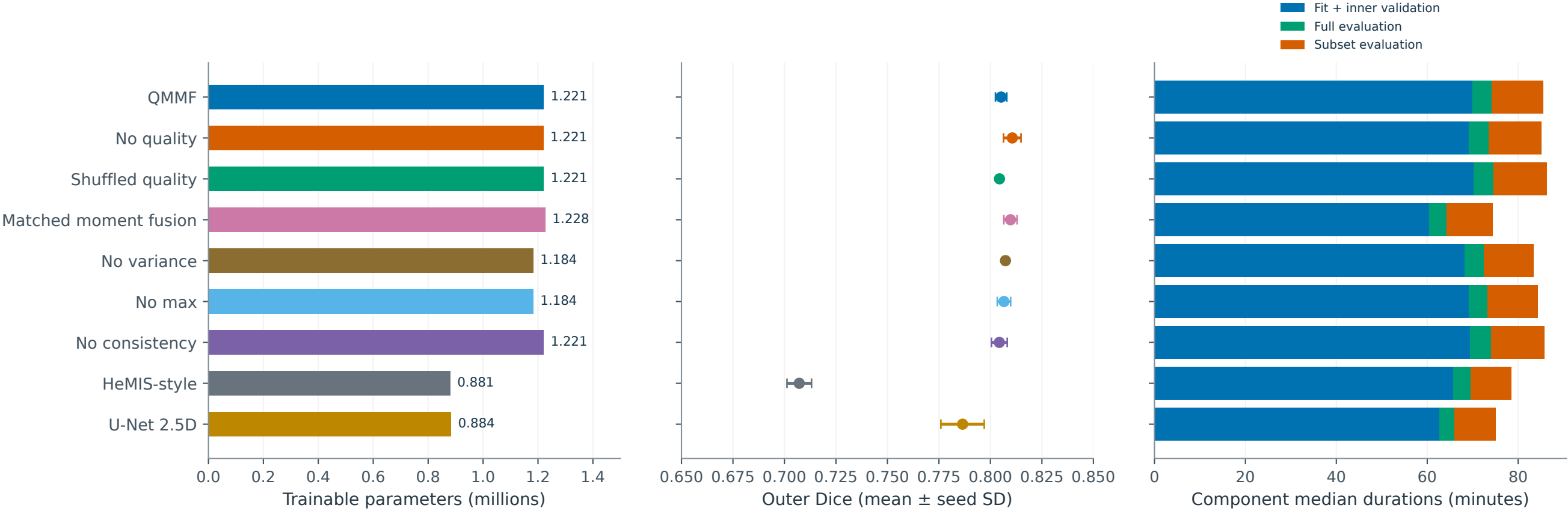
# 09 | Inner-selection learning curves

All nine variants and all three seeds; selected epochs marked with circles



# 10 | Accuracy, capacity and recorded compute

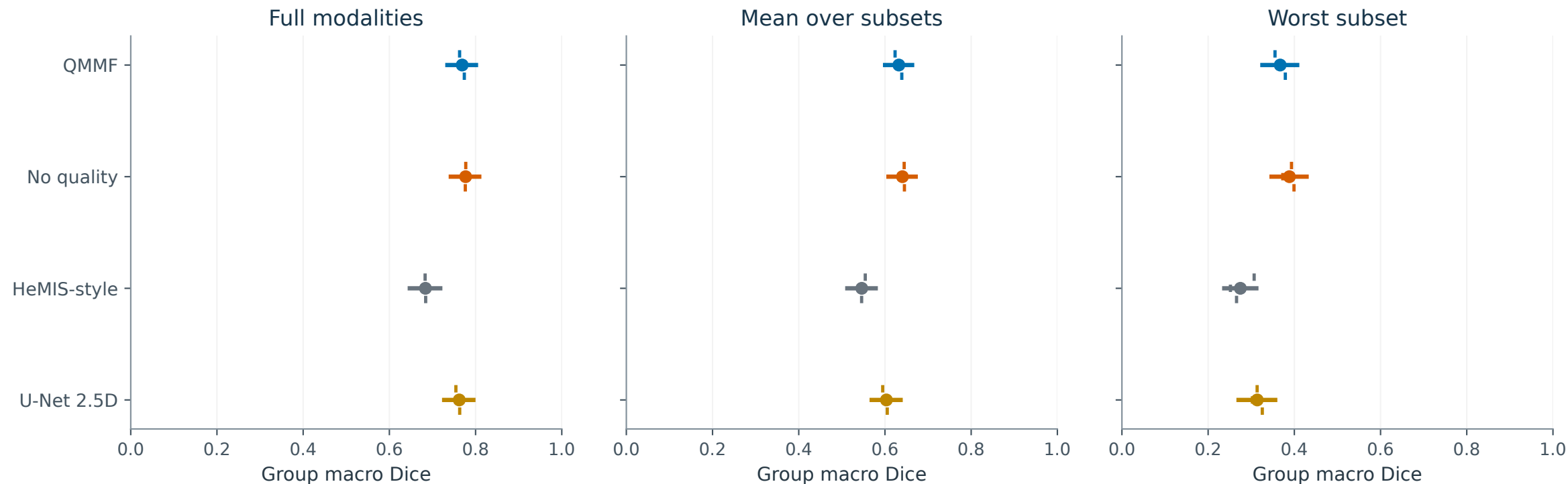
Measured resource use from the 27 main Kaggle fits; all models retained



Kaggle T4 measurements. Resource medians describe components; auxiliary-objective differences remain a confound.

# 11 | Reserved evaluation: four selected model families

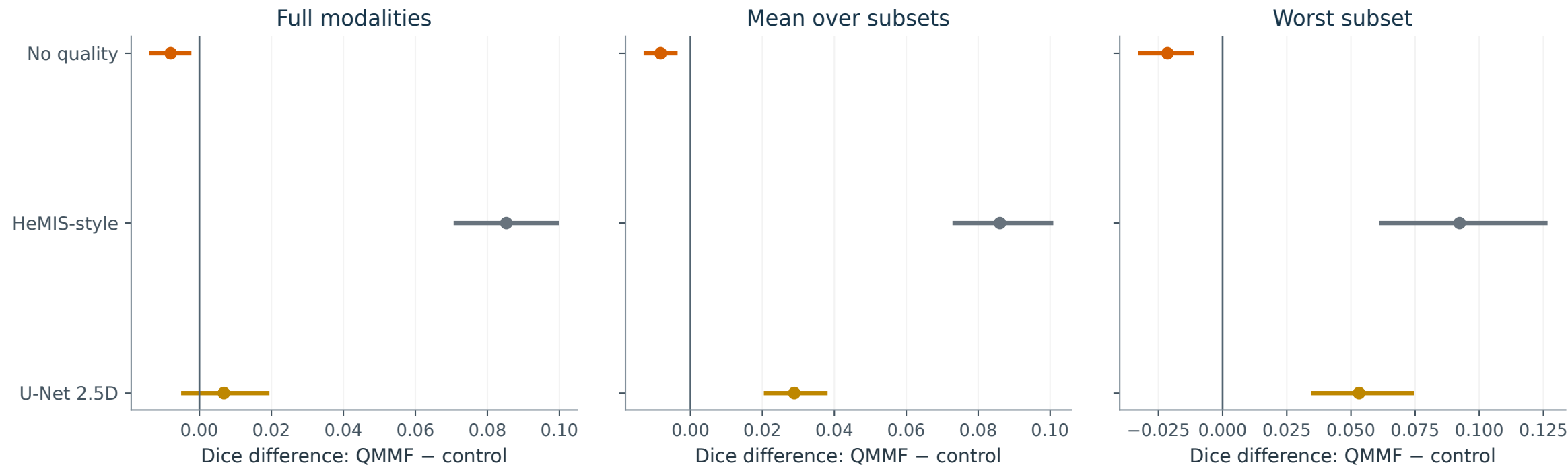
12 checkpoint evaluations; 66 cases / 51 conservative groups



Bars: 95% group-bootstrap intervals, fixed seeds. Ticks: seeds 42/43/44. Groups are not verified patients.

## 12 | Reserved paired comparisons

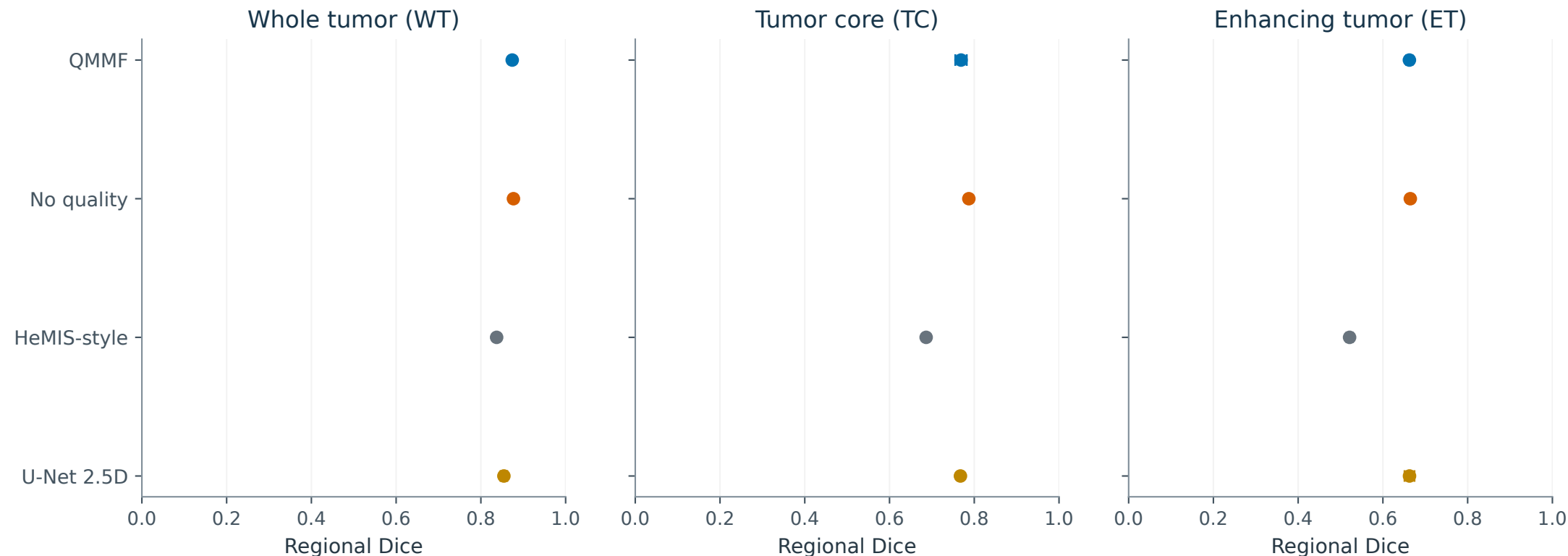
QMMF minus control; Bonferroni intervals for 3 controls per endpoint



Paired resampling unit: conservative image group. Adjustment is within endpoint, not across the entire figure gallery.

# 13 | Reserved regional segmentation

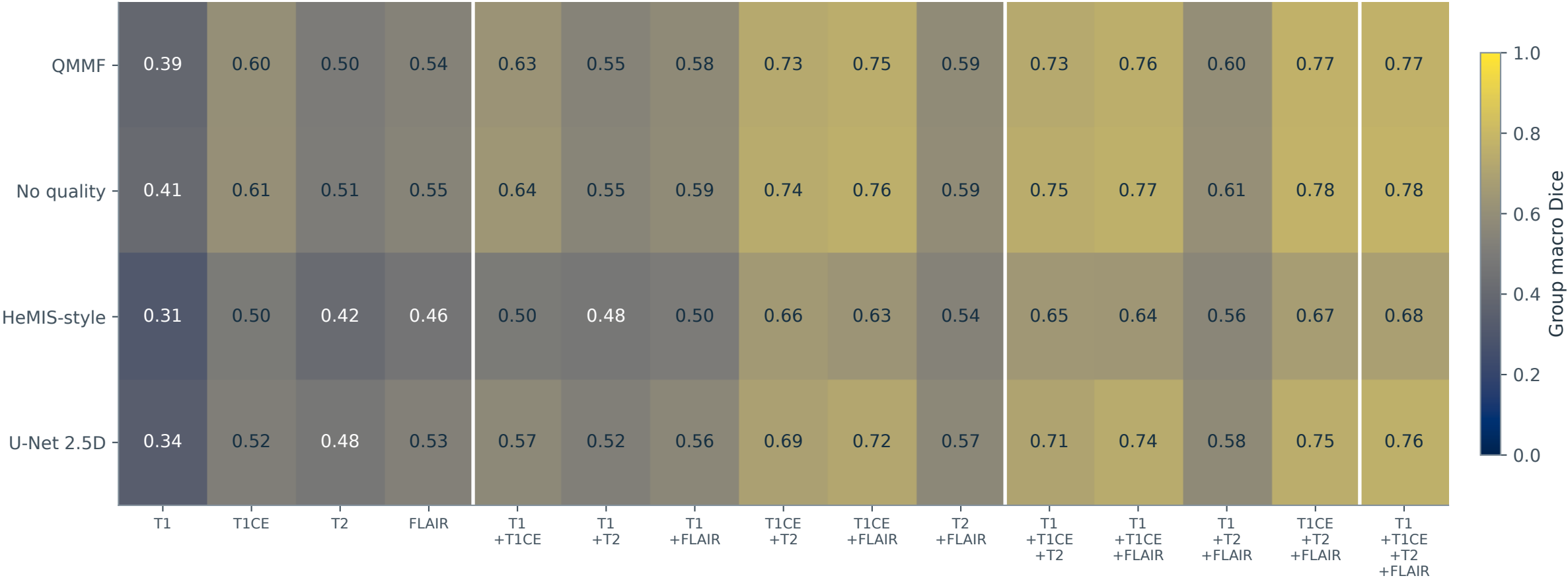
Nonempty-reference regional Dice; group aggregation followed by three-seed averaging



Bars:  $\pm 1$  seed SD. Nonempty reference only; two ET-empty cases are reported separately.

# 14 | All 15 available-modality combinations

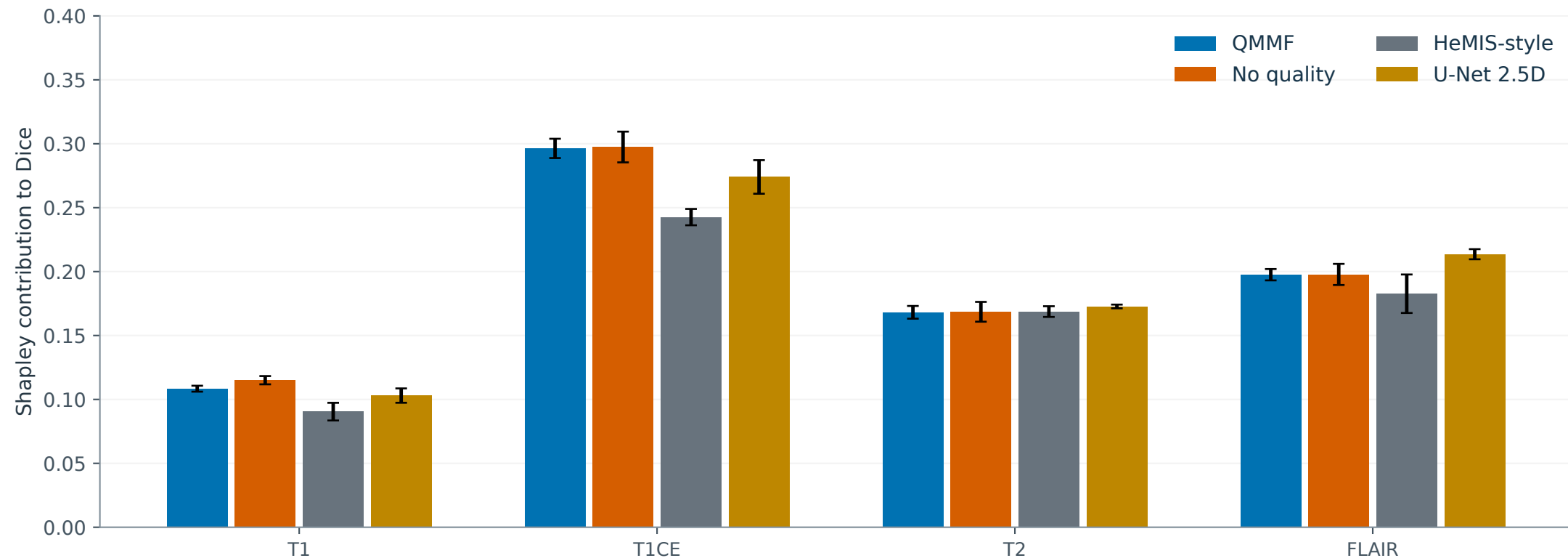
Mean group Dice over three seeds; each cell represents the same 51 reserved groups



Complete-modality preprocessing precedes masking. Cell differences are descriptive; no per-cell significance claims.

# 15 | Modality contributions under the declared subset game

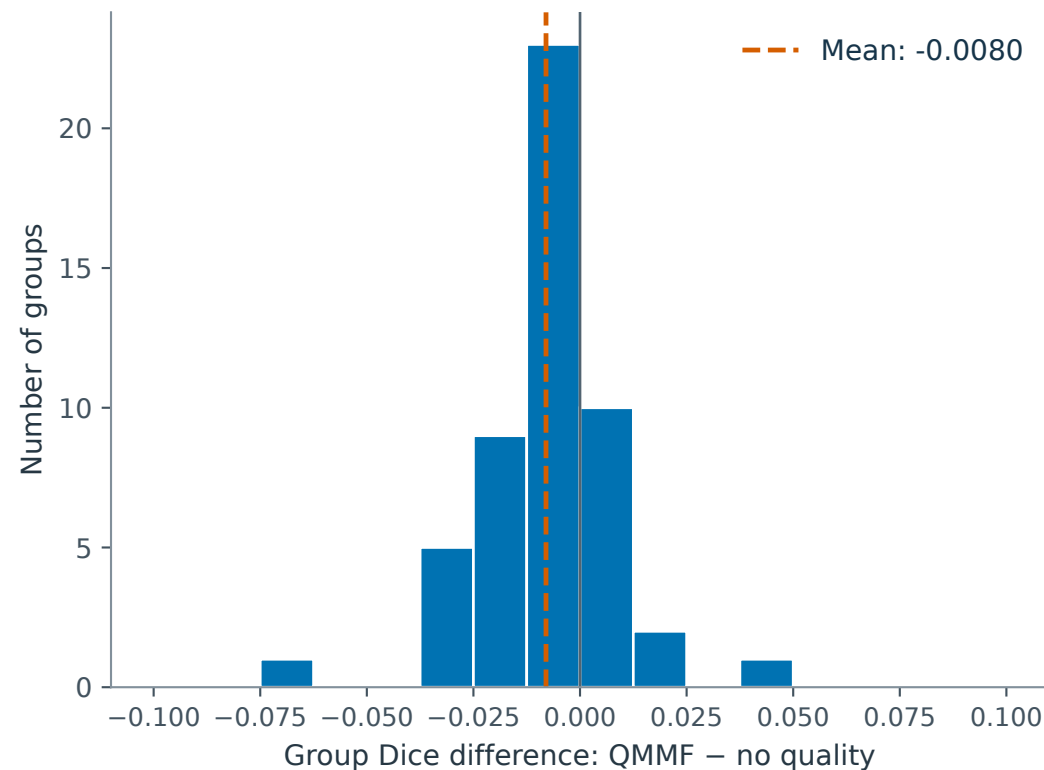
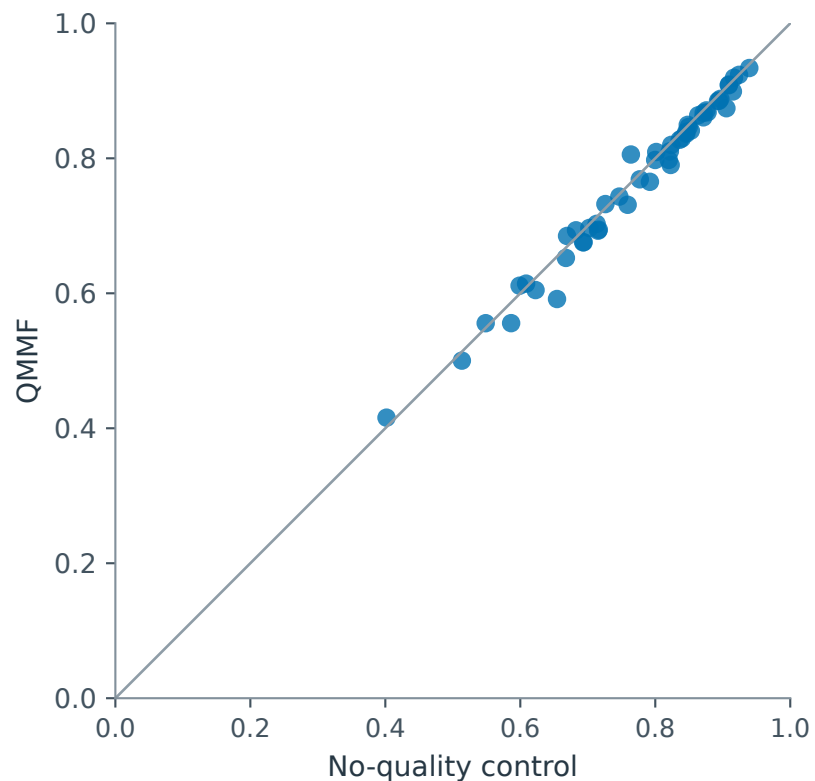
Shapley attribution of group Dice; empty-subset utility is fixed at zero



Empty utility = 0; fixed subset game. Bars:  $\pm 1$  seed SD. Attribution is descriptive, not causal.

## 16 | Group-level heterogeneity behind the average

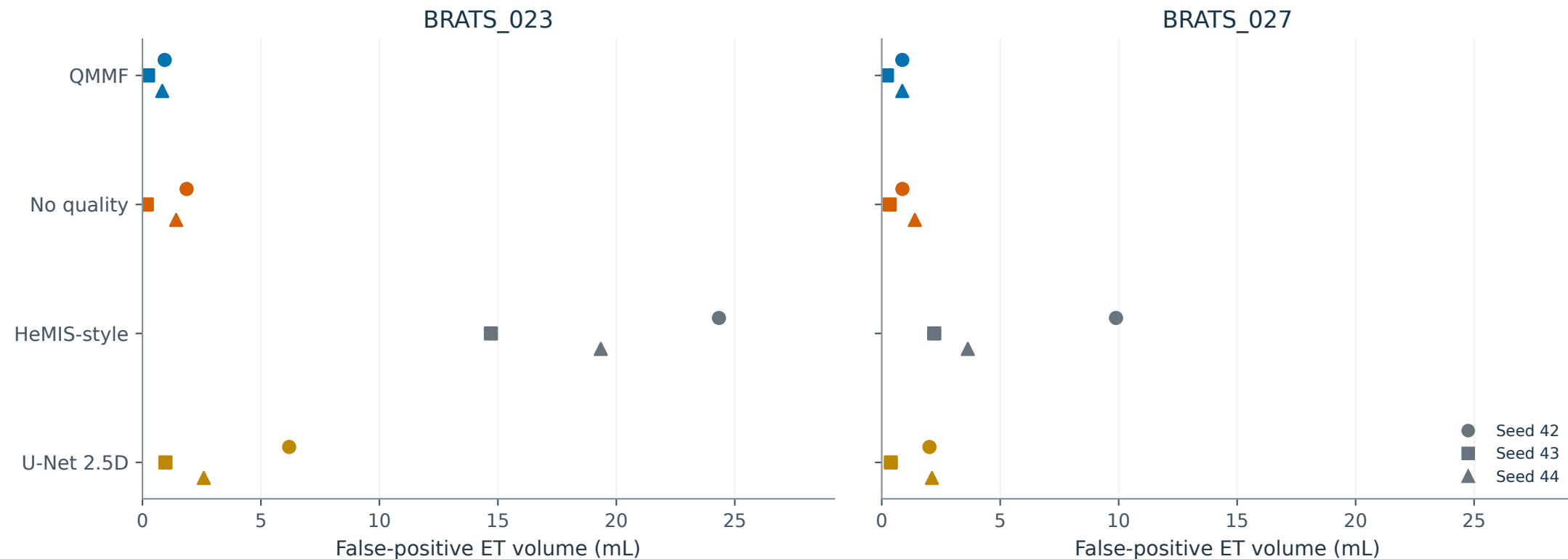
Reserved full-modality Dice, averaged over the same three fixed seeds



51 group means; each group appears once. No pooled-seed sample inflation or post hoc group selection.

# 17 | False-positive enhancing tumor on empty references

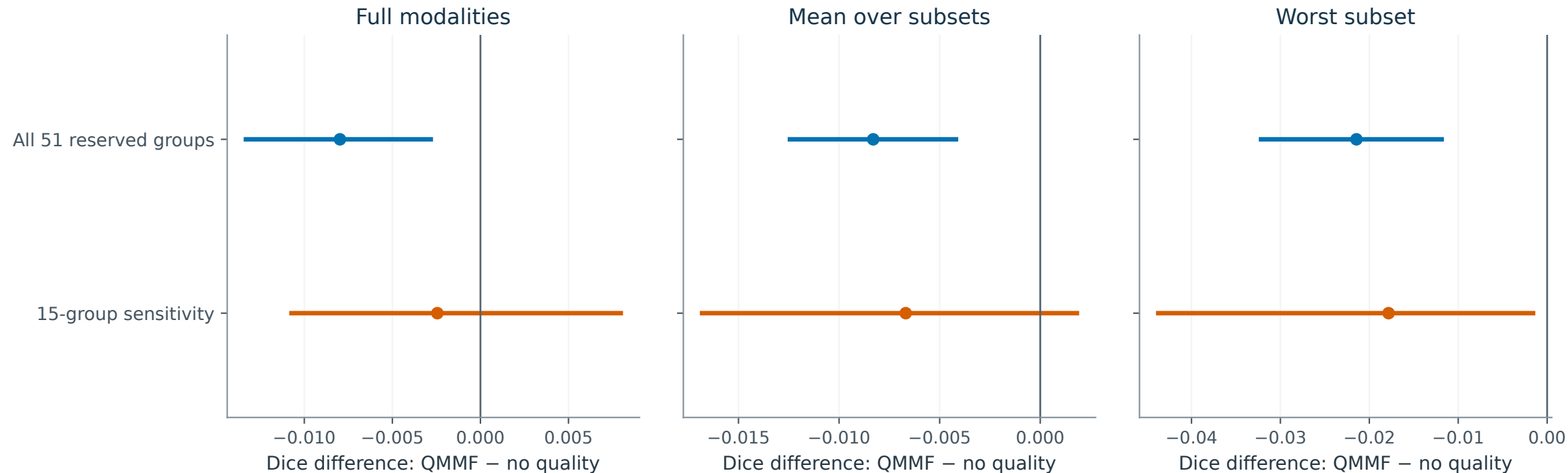
Every model and seed predicts ET in both reference-empty reserved cases



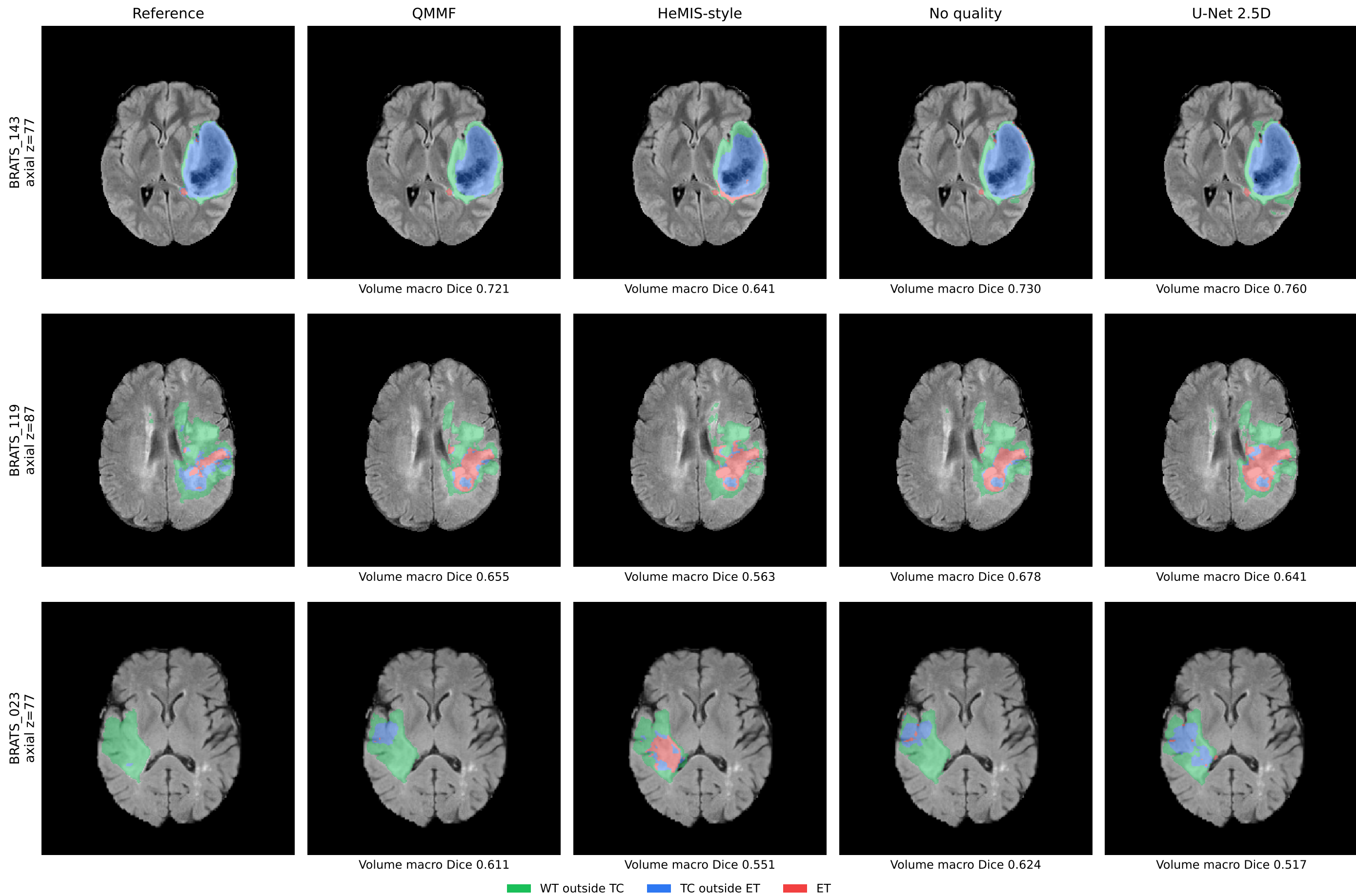
Reference ET = 0 mL in both cases. Volumes are measured outputs, not synthetic examples.

## 18 | Historical exposure sensitivity

QMMF minus no quality; retain the complete cohort and the smaller exploratory subset



The two cohorts overlap; their intervals are not an independent between-cohort test.

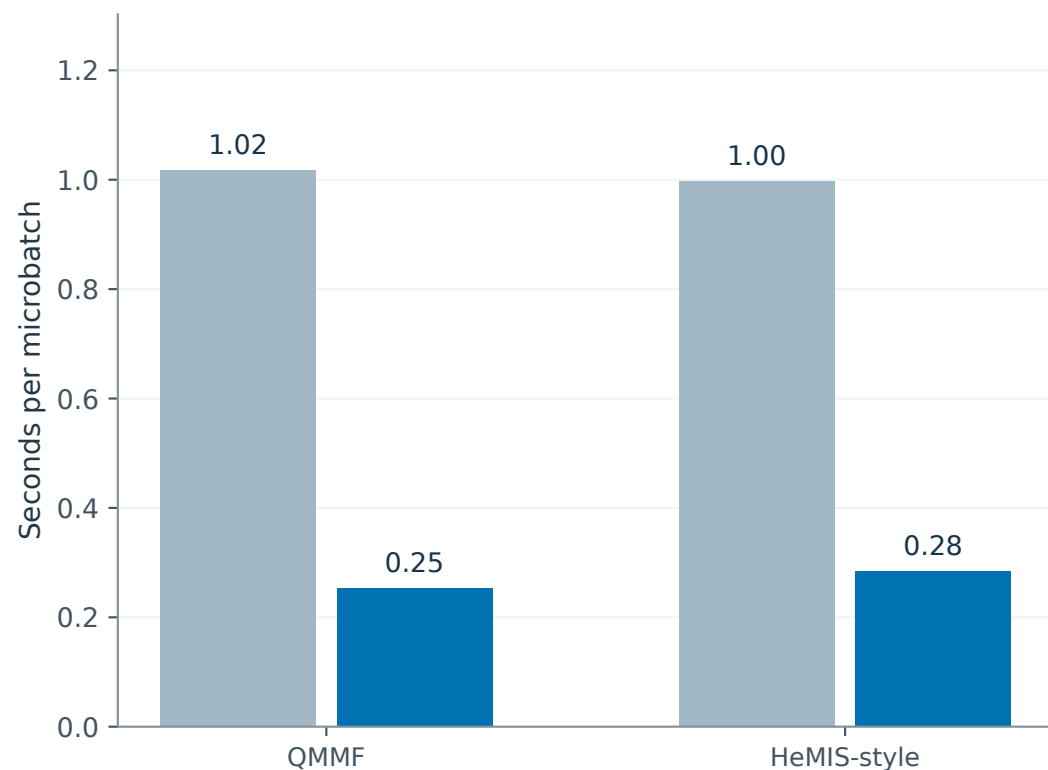
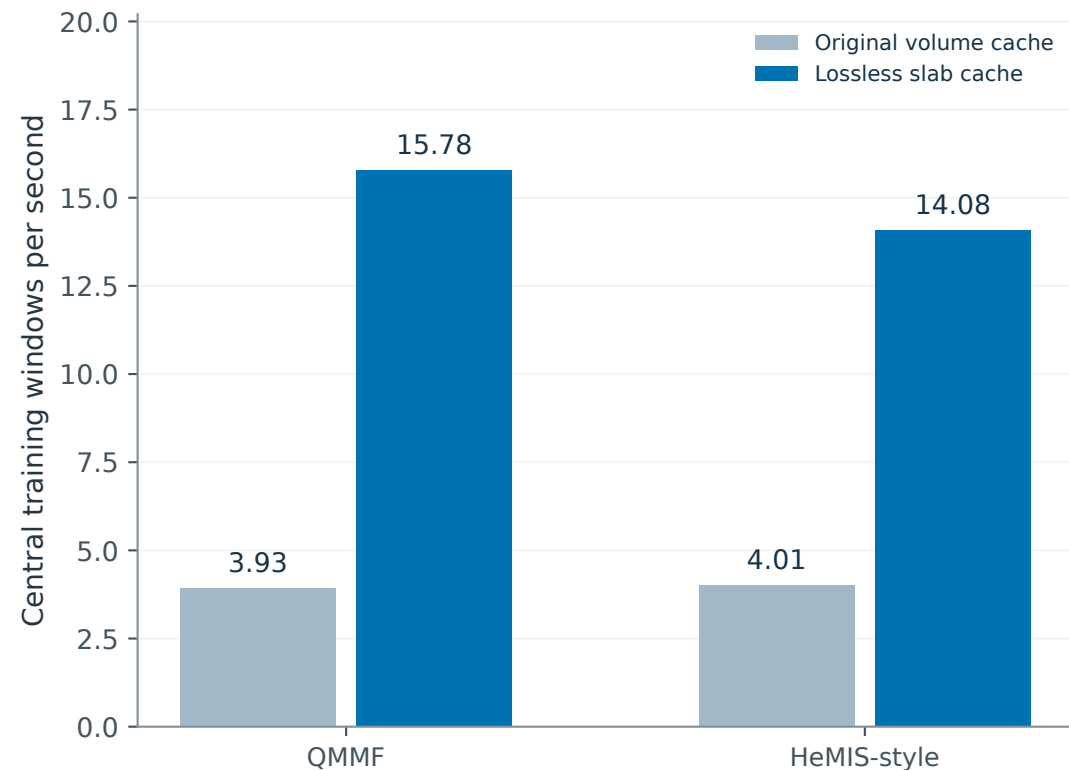


Hash-selected cases; maximum-reference-WT-area axial slices. R at viewer right; A at top.

Data: MSD/BraTS contributors, CC BY-SA 4.0. Normalized FLAIR with derived reference/prediction overlays.

## 20 | Lossless cache optimization on Kaggle

Same model and timed workload; 10 warmup + 100 measured microbatches per format



Bounded training segment only. Cache values were verified lossless; no new model fit or accuracy estimate.